

# CARBON CAPTURE

## Can we trap CO<sub>2</sub> before it heats up?

### Why CO<sub>2</sub> Matters?

Usage of fossil fuels (Releases CO<sub>2</sub>)



CO<sub>2</sub> traps heat → greenhouse effect



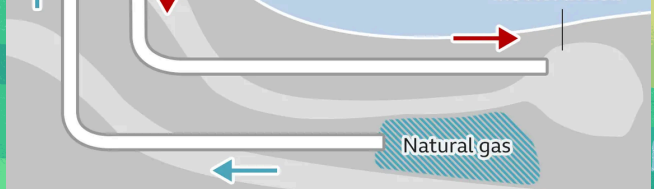
Global Warming + Climate Change

### Carbon capture and storage

1. Natural gas burned at power station

2. Carbon dioxide separated from other gases

3. Carbon dioxide stored under the North Sea

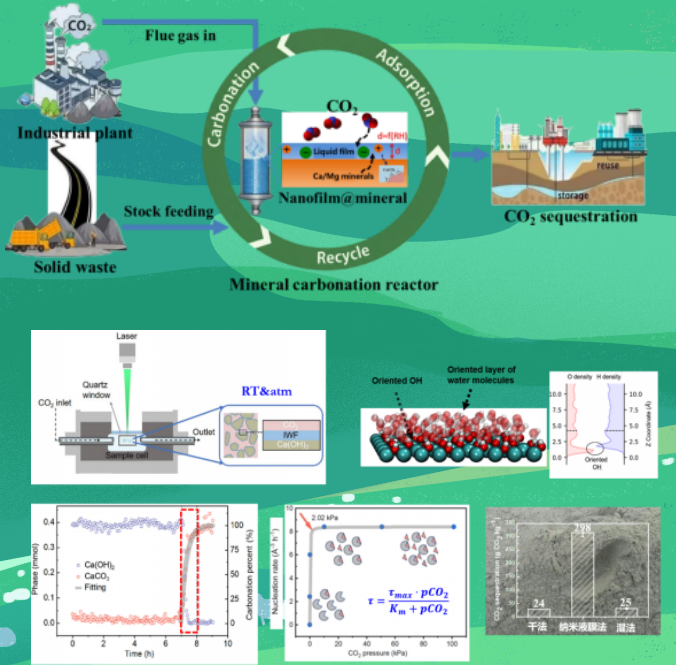


### How can we capture carbon?

Post-combustion: capture CO<sub>2</sub> after fuel is burned (Chemical Absorbers)

Pre-combustion: remove carbon before burning (Synthesis Gas)

Oxy-fuel combustion: burn fuel in pure oxygen to get concentrated CO<sub>2</sub>



### After Capture and Usage of CO<sub>2</sub>

CO<sub>2</sub> is compressed into a dense, liquid-like form and Transported via pipelines or ships

Stored in Deep saline aquifers (saltwater rock layers traps CO<sub>2</sub> for thousand years)



### Application in Real life (Usage)

- Turned into concrete, plastics, chemicals, or synthetic fuels
- Supports enhanced oil recovery
- Helps reduce net emissions
- Supports enhanced oil recovery (inject CO<sub>2</sub> into old oil wells to extract more fuel)

### Benefits of Carbon Capture

1. Climate Change Mitigation
2. Enables a Cleaner Energy Transition
3. Economic and Industrial Opportunities
4. Enhances Energy Security and System Flexibility



### Resources:

1. <https://corporate.exxonmobil.com/what-we-do/delivering-industrial-solutions/carbon-capture-and-storage/carbon-capture-storage-economic-growth-decarbonization>
2. <https://www.aramco.com/en/what-we-do/energy-innovation/advancing-energy-solutions/carbon-capture-utilization-and-storage?>
3. <https://news.ucas.ac.cn//kydt/e192701aa0894340a37ed74a77ed30db.htm>